[INCH-POUND]
A-A-59277
June 30, 1998

COMMERCIAL ITEM DESCRIPTION

OSCILLOSCOPE
(Digital Storage, 400 MHz)

The General Services Administration has authorized the use of this Commercial Item Description for all federal agencies.

1. SCOPE.

This Commercial Item Description (CID) outlines the minimum requirements for a two-channel, 400 MHz digital storage oscilloscope. The digital storage oscilloscope provided shall be commercially-available test equipment capable of being portable or modified with a rackmount kit to be included in a rackmount assembly. Only those products that meet or exceed the requirements outlined in the CID are acceptable.

2. SALIENT CHARACTERISTICS.

2.1. Class 3 equipment. Unless otherwise specified, the oscilloscope covered by this CID is categorized as Class 3 test equipment in accordance with the latest version of MIL-PRF-28800F.

2.2. Safety and environmental. The equipment shall meet all safety and environmental requirements as specified in MIL-PRF-28800F when used as bench-top or rackmount equipment.

2.3. Electrical power sources. The equipment power consumption of all components combined shall not exceed 400 Volt amps and the oscilloscope shall operate from nominal commercial military, and shipboard power sources over the following ranges: at 110/120 Volts rms (± 5 percent), single phase at frequencies of 50, 60, and 400 Hertz (Hz) and 200/240 Volts rms (± 5 percent) single phase at line frequencies of 50 and 60 Hz. The oscilloscope shall operate in accordance with the steady state conditions, transient state conditions, and interruption of power source requirements outlined in MIL-PRF-28800F.

2.4. Accessories. Accessories to be supplied with each deliverable oscilloscope shall include standard accessories as provided by the manufacturer and listed in the manufacturer's catalog or data sheet, a written tutorial/user reference manual, a written programmer reference manual, and a traceable calibration certificate.

2.4.1. Cover. The oscilloscope enclosure shall have a cover protecting the front panel controls and display in accordance with MIL-PRF-28800F.

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any data which may improve this document should be sent to: San Antonio Air Logistics Center, SA-ALC/TILDD, Building 171, Post A-16, 485 Quentin Roosevelt Rd., Kelly AFB, TX 78241-6425 by using the enclosed Form DD 1426, Standardization Document Improvement Proposal, appearing at the end of this document.

AMSC N/A FSC 6625

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2.4.2. Probes. The active or passive probes provided for both channels will permit an input signal attenuation of 1 and 10 over the full 400 MHz bandwidth and vertical accuracy. The probe attenuation shall be automatically recognized and the scale factor of the oscilloscope adjusted appropriately.

2.5. Interface. The oscilloscope shall be capable of interfacing with industry standard data interchange formats for files, formats, storage, communications, plotting and printing.

2.5.1. ANSI/IEEE-STD-488 Interface Requirements. The oscilloscope's digital interface shall control each device function including all ranges, modes, and levels except for primary power control and other non-measurement functions as required by MIL-PRF-28800F, paragraph 3.4.1.

2.5.2. Output Ports. The oscilloscope shall have RS-232 interface that complies with EIA/TIA 574 talk, talk only, and talk/listen commands and Centronics interface complying with C332-44, Feb. 1977, Rev. A, for printer and/or plotter outputs.

2.5.3. Printer Interface. The oscilloscope shall have the minimum capability of interfacing with Laser and Epson or similar style printers.

2.5.4. File Formats. The oscilloscope shall have, at the least, capability of providing, storing, and transferring data in TIFF and BMP image formats.

2.5.5. Data Format. The oscilloscope shall have the ability to process data with MathCAD™ or similar software.

2.6. Rackmount Option. If specified in the contract or order, the oscilloscope shall be delivered with an attachable rackmount kit which includes the necessary accessories and hardware to interconnect all front and rear panel input and output ports and include any filler panels to provide the preferred height and width if required. Conversion to a rack-mounted configuration shall be accomplished by use of a kit that converts the enclosure, but will not prevent return to the original configuration.

2.7. Dimensions and Weight. The oscilloscope dimensions shall permit the oscilloscope's inclusion in a standard 19-inch wide rack enclosure with a standard height of less than 6 rack units. The oscilloscope's weight, including the front cover and all accessories (excluding written manuals) shall not exceed 48 pounds.

2.8. Preconditioning Requirement. Each deliverable oscilloscope shall be capable of withstanding the environmental screening effects of burn-in and thermal cycling, with thermal cycling being accomplished in the sample testing phase.

2.9. Self-Test. The oscilloscope shall provide an automatic diagnostic self-test, in accordance with MIL-PRF-28800F paragraph 3.3.6, automatically upon start-up and/or when invoked by the operator either manually or remotely.

2.10. Extended Diagnostics. The oscilloscope shall provide extended diagnostic routines to detect internal electrical failures to the next lower level from self-test and shall be invoked by front panel controls.

2.11. Signal Path Compensation. The oscilloscope shall provide internal signal path compensation either automatically on start-up and/or when invoked remotely or by front panel controls.

2.12. Calibration. The oscilloscope shall meet the requirements for calibration outlined in MIL-PRF-28800F as well as the following:

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2.12.1. Calibration and Maintenance Adjustments. The oscilloscope shall have an auto calibration feature which ensures specified DC and timing accuracy. Changes in environmental conditions within the operating range specified for the equipment will not require additional calibration of the oscilloscope, except for internal calibration procedures and signal path compensation, to meet performance requirements.

2.13. Autoset. The oscilloscope shall provide an Autoset function that when activated, will automatically configure the vertical, horizontal, triggering, acquisition, and display controls to provide a triggered display positioned on a screen.

2.14. Temperature Requirements. The oscilloscope shall meet performance and accuracy requirements in Class 4 operating and non-operating environments. Disk operations shall be required over the range of 10 to 40 degrees C.

2.15. Humidity. The oscilloscope shall meet performance and accuracy requirements for disk drive operations in a Class 4 operating environment of 5 to 95 percent (± 5 percent) relative humidity from 10 to 30 degrees C.; to 75 percent (± 5 percent) relative humidity at 30 to 40 degrees C.; and 45 percent (± 5 percent) relative humidity above 40 degrees C.

2.16. Shock, Functional. The oscilloscope shall meet its performance and accuracy requirements in an operating environment after being subjected to a 20 G half-sine shock test.

2.17. Altitude. The maximum operating altitude with hard disk in use will be 10,000 feet, and the oscilloscope will be functional after being transported at a maximum altitude of 40,000 feet.

3.0. PERFORMANCE CHARACTERISTICS.

3.1. Warm-up Performance. The equipment shall perform to specifications after a warm-up period not to exceed thirty minutes after initiating power and all calibration and self-check routines have been accomplished.

3.2. Acquisition System. This digitizing oscilloscope shall acquire data on two channels simultaneously and all performance requirements specified herein shall be accomplished for both channels. Examples of acquisition modes include: sample, average, peak detection, envelope, high resolution, fast frame, or single sequence.

3.2.1. Sample Mode. The oscilloscope shall provide a sample mode in real time and/or repetitive sampling which may be sequential, random, interleaved or any other technique to accomplish a reconstruction of the waveform.

3.2.2. Dead Band or Dead Time Between Segments. The oscilloscope's dead band or dead between waveform captures shall be 50 µs or less.

3.2.3. Bandwidth. The oscilloscope bandwidth specifications of 400 MHz shall apply for all channels when coupled at 50 ohms or 1 Megohm. The oscilloscope bandwidth specification applies with the standard accessory probe installed.

3.2.3.1. DC Coupled Bandwidth. The oscilloscope shall have a DC response, with 50 ohms or 1 Megohm input impedance that shall not be more than -3 dB down from DC to 400 MHz for all sensitivity, and shall not be less than 350 MHz for other sensitivity ranges. The bandwidth response shall not exceed +1 dB from DC to 400 MHz.

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3.2.3.2. AC Coupled Bandwidth. The AC coupled lower frequency response at 1 Megohm input impedance shall not be more than -3 dB down at 10 Hz and the upper level frequency response shall not be more than -3 dB down at 400 MHz. The bandwidth response shall not exceed +1 dB from 10 Hz to 400 MHz.

3.2.3.3. Bandwidth Limiting. The oscilloscope shall provide the capability to limit the upper bandwidth of all channels at a single frequency within one or both over the ranges 20 to 50 MHz, or 80 to 250 MHz, and full bandwidth.

3.2.3.4. Step Recovery Settings. The oscilloscope shall recover with less than 1 percent error within 60 microseconds or less when subjected to a 2 volt step input with a rise time greater than 35 nanoseconds.

3.3. Sample Rate. The oscilloscope shall have a minimum sampling rate for real time sampling of at least 500 MS/second for each channel in operation.

3.4. Record Length. The oscilloscope shall be capable of acquiring a minimum of 50,000 samples simultaneously for each channel. Additional record lengths will be selectable as provided.

3.5. Acquisition Modes. The oscilloscope shall provide the following acquisition modes:

3.5.1. Sample. The oscilloscope shall sample data in accordance with an internal algorithm allowing data to be acquired dependent upon the time/division setting and record length selected.

3.5.2. Average. The oscilloscope shall sample data over multiple acquisitions and provide an average composite waveform of the data. The average range shall be selectable from 2 to 10,000 waveforms.

3.5.3. Peak Detection. The equipment shall capture pulses of at least 3 nanoseconds in duration for the real time sample rate.

3.5.4. Envelope Mode. The oscilloscope in a repetitive sampling mode shall capture maximum and minimum points of a series of waveform and store this data for latter display. The range for this capability shall be from 2 to 10,000 waveforms.

3.5.5. Single. The oscilloscope shall capture a single waveform in accordance with an internal algorithm allowing data to be acquired determined by the time/division setting and record length selected either manually or by a triggered input.

3.5.6. Increased Resolution. The oscilloscope shall have a mode which increases vertical resolution from 8 bits to a minimum of 10 bits of vertical resolution.

3.6. Horizontal or Time Base System. The oscilloscope shall provide the following horizontal timing functions:

3.6.1. Seconds/Division Range. The oscilloscope shall provide a seconds/division range of at least 1 nanosecond/division to 1 second/division in a 1-2-5 sequence.

3.6.2. Time Base Accuracy. The oscilloscope shall have a time base accuracy of at least ± 50 parts per million (ppm) over the temperature range of +15°C to 35°C.

3.6.3. Resolution. The oscilloscope shall have a resolution of at least 10 picoseconds.

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3.6.4. Horizontal Jitter. The oscilloscope shall have less than 50 picoseconds of horizontal jitter.

3.6.5. Delay Between Channels. The oscilloscope delay between channels shall not exceed 250 picoseconds for any channel with equal volts/division and coupling settings.

3.7. Vertical System. The oscilloscope shall function as follows:

3.7.1. Input Coupling. The oscilloscope shall provide AC coupling and DC coupling for the 1 Megohm input and DC coupling for the 50 ohm input. A ground reference shall be provided for all input channels. When in the ground reference position, the signal input shall be opened and the input circuitry to the amplifier shall be grounded.

3.7.2. Sensitivity Range. The oscilloscope shall provide a sensitivity range for all channels that is selectable in steps from 2 millivolts/division to 5 volts/division in a 1-2-5 sequence, with a X1 probe or direct input. Indicated sensitivities shall be automatically switched to display the correct scale factor when properly coded attenuator probes are attached.

3.7.3. Offset Range. The oscilloscope's offset range for all channels shall be a minimum of ± 4 divisions for all volts/division settings.

3.7.4. Position Range. The oscilloscope shall provide the capability to position each displayed trace from the center of the display to the top and bottom of the display.

3.7.5. Display Dynamic Range. The oscilloscope shall provide the capability to avoid clipping on varying signal artifacts that exceed the top and bottom display area. This may be accomplished by, but not limited to, auto setup features, display indications, warnings, increased dynamic range. These capabilities may be provided either individually or in combination.

3.7.6. Channel Isolation. The oscilloscope isolation between the input channels, with identical volts/division settings, shall be at least 40 dB at 100 MHz and at least 30 dB at 400 MHz.

3.7.7. DC Gain Accuracy. The oscilloscope DC gain accuracy shall be at least ± 2 percent of full scale at 0 volts offset.

3.7.8. Input Impedance. The oscilloscope's input impedance shall be selectable between 1 Megohm (± 1 percent) and 50 ohms (± 1 percent). The VSWR for 50 ohms input shall be less than or equal to 1.5:1 from DC to 400 MHz.

3.7.9. Maximum Input Voltage. The oscilloscope shall withstand the following requirements without damage:

3.7.9.1. One Megohm Input Impedance. The oscilloscope inputs shall withstand, without damage, up to ± 250 volts (DC plus peak AC), derated at 20 dB/decade above 10 kHz when 1 Megohm input impedance selected.

3.7.9.2. 50 Ohm Input Impedance. The oscilloscope inputs shall withstand, without damage, up to 5 volts RMS with peaks less than or equal to ± 30 volts with 50 ohms input impedance selected. The oscilloscope shall protect itself from overload of the internal 50 ohms load resistors by sensing over voltage conditions and automatically switching to 1 Megohm coupling on the overloaded input.

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3.7.9.3. Vertical Resolution. The oscilloscope shall have a minimum of 8 bits of resolution in real time sampling mode and a minimum of 10 bits resolution in repetitive sampling mode. Vertical resolution with real time sampling shall be at least 2 percent at 400 MHz.

3.7.9.4. Vertical Zoom. . The oscilloscope shall have the ability to vertically zoom up to 5X expansion or up to 50X expansion with averaging and accomplish this at a sensitivity of 50 µV/division.

3.8. Triggering System. The oscilloscope shall provide both main and delayed triggering, edge triggering, logic triggering, and pulse triggering capabilities. Edge triggering shall be fully configurable for source, slope, coupling, mode, and hold off.

3.8.1. Triggering System Bandwidth. The oscilloscope shall provide triggering over the full 400 MHz bandwidth.

3.8.2. Time Base Trigger Modes. The oscilloscope shall provide time base trigger selectable in main and delay modes.

3.9. Main Trigger . The oscilloscope shall provide the following main trigger modes:

3.9.1. Normal Mode. The oscilloscope main time base trigger shall be selectable from a vertical amplifier signal, auxiliary signal, or the power line. A waveform shall be acquired only in the presence of a trigger signal. When no trigger signal occurs, there shall be no acquisition of a waveform. Loss of the trigger shall stop any multiple acquisition sequence in progress and freeze the display, regardless of the selected time/division setting.

3.9.2. Automatic Mode. The oscilloscope shall initiate acquisition when a trigger occurs or after a certain time period (that varies based on the time/division setting) following the end of the last acquisition, whichever occurs first. A waveform shall be displayed even though the oscilloscope is not triggered. Automatic mode uses a timer that starts after a trigger event occurs. If another trigger event is not detected before the timer runs out, the oscilloscope generates another trigger.

3.9.3. Single Mode. The oscilloscope shall initiate acquisition and shall run one time or in accordance with the acquisition mode selected following rearming for each selected vertical channel and sweep combination, when triggered as in the normal mode. A front-panel control shall rearm the sweep and illuminate a reset indicator.

3.10. Delay Trigger. The oscilloscope shall have a delayed trigger mode providing delay by time or events.

3.10.1. Delay Jitter. The oscilloscope shall have a delay jitter less than or equal to 40 ppm of the time base delay setting.

3.10.2. Time Delay. The oscilloscope shall provide a trigger mode that delays the initiation of a display and timing sequence from 20 nanoseconds to 200 seconds.

3.10.3. Time Delay Accuracy. The oscilloscope delay time accuracy shall be identical to the main time base accuracy over the temperature range of +15°C to 35°C.

3.10.4. Delay By Events. The oscilloscope shall provide a trigger mode that delays the initiation of the display and timing sequence from 2 to 9,999,999 events. An event is characterized by a positive or negative transition through the trigger level as defined by the type of coupling.

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3.11. Edge Triggering. The oscilloscope shall provide edge triggering in accordance with the following:

3.11.1. Trigger Level And Slope. The oscilloscope shall provide a continuously variable trigger level control. Slope controls shall also be provided to permit triggering on any level on the positive and negative slopes of a trigger signal.

3.11.2. Trigger Level or Trigger Threshold Ranges. The upper and lower voltage limits that the trigger level or threshold can be set to are ± 4 divisions from screen center for any channel trigger source, ± 4 volts for auxiliary trigger source, and ± 250 volts for line trigger source.

3.11.3. Input Channel. The oscilloscope trigger signal is derived from any one of the primary inputs or trigger channel. If the trigger source is one of the primary channels selected for display, you can view the trigger source or trigger from an input source not being displayed.

3.11.4. AC Line Voltage. The oscilloscope trigger signal is derived from the AC line voltage. AC line is available as a source only for the main time base trigger.

3.11.5. External Trigger. The oscilloscope shall provide an external trigger input.

3.11.5.1. External Trigger Level Range. The scope shall have a trigger level range greater than ± 500 millivolts.

3.11.5.2. External Trigger Level Edge Sensitivity. The oscilloscope shall trigger on a minimum of 400 millivolts up to 50 MHz and a minimum of 750 mV up to 100 MHz.

3.11.5.3. External Trigger Level Maximum Input. The oscilloscope shall accept external inputs up to ± 20volts (dc + peak ac).

3.12.5.4. External Trigger Bandwidth. The oscilloscope external bandwidth shall be equal to or exceed the normal triggering bandwidth of the equipment.

3.11.6. Edge Type Trigger Coupling. The oscilloscope shall provide the following trigger coupling modes:

3.11.6.1. Direct Current Trigger (DC). The oscilloscope shall trigger on both AC and DC signal components when signal levels exceed 0.5 divisions to 50 MHz and 1.5 divisions to 400 MHz.

3.11.6.2. Alternating Current Trigger (AC). The oscilloscope shall trigger on the AC signal components above approximately 60 Hz when signal levels exceed 0.5 divisions to 50 MHz and 1.5 divisions to 400 MHz.

3.11.6.3. High Frequency Rejection. The oscilloscope shall trigger on signals at 0.5 divisions from DC to approximately 30 kHz.

3.11.6.4. Low Frequency Rejection. The oscilloscope shall trigger when signals exceed 0.5 divisions above 80 kHz to 50 MHz and 1.5 divisions to 400 MHz.

3.11.7. Trigger Position. The position of the trigger in the waveform record shall be selectable in 1 percent increments to allow 0 to 100 percent of the data to be acquired.

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3.11.7.1. Trigger Jitter. The oscilloscope trigger jitter shall not exceed 100 picoseconds at full bandwidth and the fastest sweep setting.

3.11.8. Exclusion Triggering. The oscilloscope shall trigger on a signal or period that fall outside two limits which are settable from 2.5 nanoseconds to 20 seconds.

3.11.9. Pulse Triggering. The oscilloscope shall provide triggering on pulse edges with the minimum characteristics described herein.

3.11.9.1. Glitch Triggering. The oscilloscope shall trigger or reject signals on the rising or falling edges of pulses with widths at least over the range of 5 nanoseconds to 1 second.

3.11.9.2. Glitch Triggering Time Accuracy. The oscilloscope shall have a minimum time accuracy for pulse width triggering for a time range of 5 nanoseconds to 1 microsecond, of ± (20 percent of setting + 1.5 nanoseconds) and for a time range greater than 1 microsecond and less than or equal to 1 second; accuracy shall be ± .5 percent.

3.11.9.3. Pulse Width Triggering. The oscilloscope shall trigger on a positive or negative pulse within or not within the selectable time limits of at least 2.5 nanoseconds to 1 second.

3.11.9.3.1. Time Accuracy For Pulse Width Triggering. The oscilloscope shall have a minimum time accuracy for pulse width triggering for a time range of 2.5 nanoseconds to 1 microsecond, of ± (20 percent of setting + 1.5 nanoseconds), and for a time range greater than 1 microsecond; accuracy shall be ± .5 percent measured at 50 percent.

3.11.10. Time-out Triggering. The oscilloscope shall trigger when a pulse fails to complete a transition from a high to low state and return to the original state within the time range of at least 2.5 nanoseconds to 20 seconds.

3.11.11. Video Triggering. The oscilloscope shall provide video triggering in PAL, NTSC, SECAM, and non-standard video formats in monochrome or color.

3.11.11.1. Video Trigger Selection. The oscilloscope shall trigger on the TV input signal referenced to line or field number or both.

3.11.11.2. Video Trigger Sensitivity. The oscilloscope shall trigger on .8 divisions or less of composite SYNC signal.

3.11.11.3. Video Synchronization. The oscilloscope shall trigger at least on falling horizontal synchronization signals.

3.11.12. Non-Standard Video Triggering. The oscilloscope shall provide a triggering capability for non-standard video by providing the ability of defining repetition rate, number of lines, number of fields, and vertical synchronous timing to initiate a trigger action.

3.11.13. Enhanced Video Resolution. The oscilloscope shall be able to display and analyze at least 1,500 lines per inch of video signals.

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3.11.13.1. Magnification. The oscilloscope shall provide a horizontal magnification adjustable over a minimum range of 1 to 1000. The oscilloscope shall position waveforms both vertically and horizontally and read out on screen of the vertical and horizontal scaling factors.

3.12. Measurement System Features. The oscilloscope shall provide the following measurement capabilities:

3.12.1. Automated Measurements. The oscilloscope shall provide automated measurements defined by IEEE-STD-181-1977. The oscilloscope measure function will automatically extract parameters from any signal input to the scope. Four or more of the parameters can be displayed on screen and made available for transfer over the IEEE-488 interface bus. The parameters measured shall include, but are not limited to: period, frequency, width, rise time, fall time, duty cycle, delay, positive overshoot, negative overshoot, maximum, minimum, peak-to-peak, amplitude, root mean square (rms), propagation delay, burst, mean and duration.

3.12.2. Automated Measurement Accuracy Voltage. The scope shall have a minimum accuracy of ± 2% for voltage related measurements, for single and dual point measurements with the signal at 50% or higher of full scale.

3.12.3. Automated Measurement Accuracy Time. The oscilloscope shall have minimum accuracy of ± 1 percent time-related measurements for single and dual point measurements.

3.12.4. Cursor Measurements. The oscilloscope shall have the capability to make single and dual cursor measurements along both the vertical and horizontal axis with and without tracking. Cursors shall be provided for making vertical and horizontal measurements on the displayed waveforms. The cursors shall have the ability to measure absolute and delta voltage between the positions of the horizontal bar cursors and absolute and delta time in seconds or Hz for 1/time between the positions of the vertical bar cursors. The time and voltage related values represented by cursor positions shall be displayed on screen and available for transfer over the IEEE-488 interface.

3.12.4.1. Cursor Measurement Accuracy Voltage. The oscilloscope shall have a minimum accuracy of ± 2 percent of full scale for voltage related measurements when utilizing single cursors and ± 4.0% when using dual cursors.

3.12.4.2. Cursor Measurement Accuracy Time. The oscilloscope shall have a minimum accuracy of ± 1 percent of full scale for time related measurements when utilizing either single or dual cursors.

3.12.4.3. Cursor Readout. The oscilloscope shall provide a cursor readout selectable by the user to display cursor measurement in either voltage, time, or frequency or any combination of these measurements. Dual cursor measurements shall provide a readout of the relative positions of the cursors and the difference.

3.12.5. Math Functions. The oscilloscope shall provide math waveform operations of addition, subtraction, inversion, and multiplication of any two waveforms in any combination from the current displayed waveforms or stored waveforms.

3.13. Display, Controls, & Adjustments. This oscilloscope shall have the following display, controls & adjustments:

3.13.1. Display. A display or visual device capable of displaying both waveforms in normal ambient room light in sufficient detail for direct measurements on screen without automated measurements or cursors. The minimum viewing area of the display shall be at least 10 centimeters (cm) horizontally and 8 cm vertically.

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3.13.2. Display Style. The oscilloscope shall provide waveform displays in sin (x)/x, linear, dots, variable persistence, and infinite persistence styles.

3.13.3. Trace Convergence. The oscilloscope shall have the capability to discriminate and display where multiple waveforms converge.

3.14. Display Intensity. The oscilloscope shall provide the capability to control overall intensity, text/graticule intensity, and waveform intensity.

3.15. Display Persistence. The oscilloscope shall provide both infinite and variable persistence. The variable persistence range shall be over the range of 500 ms to 10 seconds with at least 50 ms resolution.

3.16. Display Gray Scaling. The oscilloscope shall provide the ability to accomplish gray scaling of at least 16 discernable intensity levels.

3.17. Display Graticule Size and Markings. The oscilloscope performance shall be over a minimum 8 X 10 division graticule in the waveform display area. The cardinal axes shall be ruled in 0.2 minor divisions.

3.18. Display Graticule Types. The oscilloscope shall provide the capability of selecting full screen, single window, dual window, XY display, grid, or crosshair displays.

3.18.1. Waveform Display Area. The oscilloscope waveform display area shall have at least 640 horizontal points by at least 480 vertical points.

3.18.2. Screen Data Displayed. Horizontal and vertical scaling factors for each channel, main and delayed time bases, trigger sources, trigger level, trigger type, sample rate, mode, at least four measurements, and a representation of memory length with trigger pint indicated.

3.19. Format. The oscilloscope shall have both YT and XY axis formats.

3.20. "Fit to" and "Fill" The Screen. The oscilloscope shall have the ability to display the entire memory acquisition on the screen.

3.21. X-Y Operation. The oscilloscope shall provide X-Y operation.

3.21.1. X-Axis and Impedance Range. The scope shall have an X-axis deflection factor, variable range, and accuracy as well as an input impedance range defined for channels 1 and 2.

3.21.2. X and Y Phase Difference. The oscilloscope phase difference between X and Y shall be less than 1° from DC to 1 MHz and less than 2° from 1 MHz to 2 MHz.

3.22. Waveform/Setup Storage. The oscilloscope shall provide internal waveform storage to store waveforms, reference waveforms, front panel setups, and calibration data.

3.22.1. Waveform and Set-Up Storage. The oscilloscope shall provide internal memory to store a minimum of four waveforms at maximum memory storage and sufficient internal memory to store four front panel set-ups.

3.23. Back Up Battery. The oscilloscope shall have a rechargeable back up battery with an internal memory of at least two years. The equipment shall have the capability to store front panel settings and to retain this data

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for a minimum of two years. If the equipment uses a Nickel-Cadimum (Ni-Cad) battery, there shall be means by which the operator and/or internal circuitry can deep discharge the battery to prevent "memory effect."

3.24. Memory Data Transfer. The oscilloscope shall provide the capability to save any waveform or setup data to any applicable reference memory location, to transfer setup information and/or waveforms into a reference memory location via the GPIB interface, and to transfer setup and waveform data from the floppy disk drive.

3.25. Mass Storage. The oscilloscope shall provide mass storage providing additional storage of reference waveforms, setups, data files and image files. The oscilloscope may have a floppy disk in a 3.5 inch size, 1.4 Megabytes storage, and in a DOS format.

3.26. Alternate Mass Storage Devices. The oscilloscope may use a PCMCIA I/II/III flash memory card or other storage device with 1.4 Megabytes or greater capacity and DOS format.

3.27. Mass Storage Operational Restraints. The oscilloscope shall have the ability store, archive, retrieve, and any other storage operation over a temperature range of +10°C to +40°C.

4. REGULATORY PROVISIONS.

4.1. In accordance with Section 23.403 of the Federal Acquisition Regulation, the federal government's policy is to acquire items composed of the highest percentage of recoverable materials practicable consistent with maintaining a satisfactory level of competition and without adversely affecting performance requirements or exposing the supplier's employees to undue hazards from the recovered materials.

5. QUALITY ASSURANCE PROVISIONS.

5.1. Product Conformance. The oscilloscope provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The federal government reserves the right to require proof of such conformance.

5.2. Submitted Products. The equipment submitted for bid sample testing shall be representative of the commercial off-the-shelf product. All features, performance, and environmental protection of the product tested during bid sample testing shall be provided under any resulting contract or purchase order.

5.3. Year 2000 Compliance. This oscilloscope shall accurately process date ranges and time data from, into, and between the 20th and 21st Centuries and specifically, the years 1999, 2000, 2001, and leap year calculations. This compliance includes calculating, comparing, and sequencing date/time data. The oscilloscope shall also accurately process date and/or time data if such data is received or exchanged from other equipment or sources.

6. PACKAGING.

6.1. Preservation, Packaging, Labeling, And Marking. The preservation, packaging, packing, labeling, and marking shall be as specified in the contract or order. The oscilloscope shall have an identification plate permanently affixed in a readable place which includes the information outlined in MIL-PRF-28800F, paragraph 3.11.2.

7. NOTES.

7.1. Metric Products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the tolerances specified in conversion tables

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contained in the latest version of Federal Standard No. 376 and all other requirements of this CID are met. If the manufacturer elects to use metric dimensions as the basis upon which to derive or base a mechanical feature, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

7.2. Source of Document Publications. Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Defense Automated Printing Service, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094. Phone 215/697-2179.

7.3. Possible Acceptable Products.

Manufacturer	CAGE	Model
LeCroy 700 Chestnut Ridge Road Chestnut, NY 10977-6499	55716	9314A
Tektronix, Inc. P. O. Box 500 Beaverton, OR 97077	80009	TDS 520C

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c. ADDRESS *(Include Zip Code)* SA-ALC/TILDD (MS SYLVIA A. REYES) 485 QUENTIN ROOSEVELT RD, B171 KELLY AFB, TX 78241-6425	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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